

Generalized Linear Models

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1 Introduction

Suppose we have the following experiment:

- We have data on 79 cervical cancer patients, and we want to study the association between histology (low v.s high) and the risk factors (neovasculation, pulsatility, endometrium height).
- We would like to use regression methods by coding histology as 0 (low) and 1 (high) and using those 3 factors as covariates.

There are many potential issues with this experiment. OLS is genius, but potentially inefficient or awkward to work with. One alternative is to first observe that the response variable Y_i is Bernoulli (0 or 1), so we first assume

$$\mathbb{P}(Y_i = 1|\mathbf{x}_i) = p(x_1, x_2, x_3|\beta),$$

where $p_i(\beta) \equiv p(x_1, x_2, x_3|\beta)$ is any function of the three covariates and an unknown parameter vector β . The likelihood of the n observations is

$$L(y_1, \dots, y_n; \beta) = \prod_{i=1}^n p_i(\beta)^{y_i} \cdot (1 - p_i(\beta))^{1-y_i},$$

and the MLE is given by

$$\hat{\beta} = \arg \max_{\beta} \prod_{i=1}^n p_i(\beta)^{y_i} \cdot (1 - p_i(\beta))^{1-y_i}.$$

For a particular choice of p , we can then try to maximize this either analytically or numerically. The first choice for p is the CDF of a $\mathcal{N}(0, 1)$ random variable:

$$p_i = \Phi(\hat{z}) = \int_{-\infty}^{\hat{z}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^2\right) du,$$

and $\Phi^{-1}(p_i)$ is often referred to as the probit function. The standard approach is to then assume

$$\Phi^{-1}(p_i) = \beta_0 + \beta_1 x_i.$$

The disadvantages for this model is that is it computationally challenging, also not an obvious interpretation of parameters for even a univariate model.

An alternative approach is to use the *logit*:

$$p_i = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}.$$

Under this model, we have

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x.$$

2 Logistic Regression

2.1 Understanding logistic regression

In the logistic regression model,

$$p_i = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}.$$

Under this model, we have

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x.$$

We call $\frac{p_i}{1-p_i}$ the *odds* of $Y_i = 1$. β_0 is the *log-odds* for an observation with $x = 0$, $\exp(\beta_0)$ would be the odds ratio for such an observation. β_1 is thus the increase in the log-odds of a positive response for a one unit increase in x , $\exp(\beta_1)$ is the *relative* increase in odds.

We can center the covariate x by using $x_c = x - \bar{x}$ which yields

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0^* + (x_i - \bar{x})\beta_1, \quad \beta_0^* = \beta_0 + \bar{x}\beta_1.$$

Take two observations, indexed i, j with covariate x and $x + 1$, then the risk ratio is

$$\begin{aligned} \frac{\mathbb{P}(X_j = 1)}{\mathbb{P}(X_i = 1)} &= \frac{p_j}{p_i} \\ &= \frac{\exp(\beta_0 + (x + 1)\beta_1)}{1 + \exp(\beta_0 + (x + 1)\beta_1)} \cdot \frac{1 + \exp(\beta_0 + x\beta_1)}{\exp(\beta_0 + x\beta_1)} \\ &= \exp(\beta_1) \cdot \frac{1 + \exp(\beta_0 + x\beta_1)}{1 + \exp(\beta_0 + (x + 1)\beta_1)} \\ &\approx \exp(\beta_1), \end{aligned}$$

when p_i, p_j are small. Therefore, for small values of p_i (when the response probability is small), then the odds ratio $\exp(\beta_1)$ can be interpreted as an approximate relative increase in the probability of response for a one unit increase in x . In many applications, this is true.

2.2 Asymptotic Normality of the MLE

In a univariate logistic regression model, recall that

$$p_i = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}.$$

The response Y_i is binary: 0, 1, so the likelihood of the sample is given by

$$L(\beta_0, \beta_1) = \prod_{i=1}^n p_i(\beta_0, \beta_1, x)^{y_i} \cdot (1 - p_i(\beta_0, \beta_1, x))^{1-y_i},$$

with the log-likelihood given by

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^n y_i \log p(\beta_0, \beta_1, x) + (1 - y_i) \log(1 - p(\beta_0, \beta_1, x)). \quad (2.1)$$

Suppose we can maximize the log-likelihood defined in (2.1) and get the maximum likelihood estimator $\widehat{\boldsymbol{\theta}} = (\widehat{\beta}_0, \widehat{\beta}_1)$:

$$(\widehat{\beta}_0, \widehat{\beta}_1) := \arg \max_{\beta_0, \beta_1} \sum_{i=1}^n y_i \log p(\beta_0, \beta_1, x) + (1 - y_i) \log(1 - p(\beta_0, \beta_1, x)).$$

The MLE estimate $\widehat{\boldsymbol{\theta}} = (\widehat{\beta}_0, \widehat{\beta}_1)$ can then be the estimated coefficients for univariate logistic regression. In asymptotic theory, the maximum likelihood estimator is asymptotically normal. We first introduce the regularity conditions:

Definition 1 (Regularity Conditions). *In a random sample $X_1, \dots, X_n \sim f(x, \theta)$ where θ is one-dimensional, we give the following regularity conditions:*

- (i) *The family $\{f(x, \theta) : \theta \in \Theta \subset \mathbb{R}\}$ has a common support that does not depend on θ .*
- (ii) *$\frac{d}{d\theta} \log f(x, \theta)$ always exists.*
- (iii) *For any statistic $h(\mathbf{X}) \in L^1$,*

$$\frac{d}{d\theta} \int_{\mathcal{S}} h(\mathbf{x}) f(\mathbf{x}, \theta) d\mathbf{x} = \int_{\mathcal{S}} h(\mathbf{x}) \frac{d}{d\theta} f(\mathbf{x}, \theta) d\mathbf{x}$$

Note that the likelihood cannot (except in certain cases) be maximized analytically, so one must resort to numerical methods. But we will talk about that later. Assume for now that we can obtain ML estimate $\widehat{\boldsymbol{\theta}}$ somehow, and let the true distribution that generates the data be $q(y)$.

Theorem 1. *If there exists a parameter vector $\boldsymbol{\theta}_0$ such that $p_1(\boldsymbol{\theta}_0) = q(y_i)$ for all observations, where q is the true distribution. Then the MLE of $\boldsymbol{\theta}$ converges to $\boldsymbol{\theta}_0$ in probability, and moreover*

$$\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathcal{J}^{-1}(\boldsymbol{\theta}_0)),$$

where

$$(\mathcal{J}(\boldsymbol{\theta}_0))_{ij} = -\mathbb{E} \left[\frac{\partial^2 \ell(\boldsymbol{\theta}, y)}{\partial \theta_i \partial \theta_j} \right] \bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}.$$

We will prove the theorem in a univariate setting. By i.i.d assumption, we can write the log-likelihood as

$$\ell_n(\theta, y) = \sum_{i=1}^n \ell(\theta, y_i).$$

Let $\dot{\ell}_n(\theta, y)$ be the first derivative of the log-likelihood with respect to θ for n i.i.d observations. We then do a Taylor expansion for $\ell_n(\theta, y)$ around θ_0 :

$$\dot{\ell}_n(\theta, y) = \dot{\ell}_n(\theta_0, y) + (\theta - \theta_0) \cdot \ddot{\ell}_n(\theta_0, y) + \frac{(\theta - \theta_0)^2}{2} \cdot \ddot{\ell}_n(\theta^*, y), \quad \theta^* \text{ is between } \theta, \theta_0.$$

Since $\widehat{\theta}_n$ is the MLE, so $\dot{\ell}_n(\widehat{\theta}_n, y) = 0$, hence

$$0 = \dot{\ell}_n(\theta_0, y) + (\widehat{\theta}_n - \theta_0) \cdot \ddot{\ell}_n(\theta_0, y) + \frac{(\widehat{\theta}_n - \theta_0)^2}{2} \cdot \ddot{\ell}_n(\theta^*, y),$$

which implies

$$-\dot{\ell}_n(\theta_0, y) = (\hat{\theta}_n - \theta_0) \cdot \ddot{\ell}_n(\theta_0, y) + \frac{(\hat{\theta}_n - \theta_0)^2}{2} \cdot \ddot{\ell}_n(\theta^*, y),$$

hence

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{\dot{\ell}_n(\theta_0, y)/\sqrt{n}}{-\frac{1}{n}\ddot{\ell}_n(\theta_0, y) - \frac{(\hat{\theta}_n - \theta_0)}{2n} \cdot \ddot{\ell}_n(\theta^*, y)}. \quad (2.2)$$

To understand better, we need to know the following properties of $\dot{\ell}(\theta, y_i)$:

- $\mathbb{E}[\dot{\ell}(\theta_0, y_i)] = 0$;
- $-\mathbb{E}[\ddot{\ell}(\theta_0, y_i)] = \mathbb{E}[\dot{\ell}(\theta_0, y_i)^2]$.

Those are known as the *Bartlett's identities*.

- First Bartlett Identity:

$$\mathbb{E} \left\{ \frac{d}{d\theta} \log p_\theta(\mathbf{x}) \right\} = 0, \forall \theta \in \Theta;$$

- Second Bartlett Identity:

$$\mathbb{E} \left\{ \left(\frac{d}{d\theta} \log p_\theta(\mathbf{x}) \right)^2 \right\} = -\mathbb{E} \left\{ \frac{d^2}{d\theta^2} \log p_\theta(\mathbf{x}) \right\}.$$

Then in (2.2) we have

$$\begin{aligned} \sqrt{n}(\hat{\theta}_n - \theta_0) &= \frac{\dot{\ell}_n(\theta_0, y)/\sqrt{n}}{-\frac{1}{n}\ddot{\ell}_n(\theta_0, y) - \frac{(\hat{\theta}_n - \theta_0)}{2n} \cdot \ddot{\ell}_n(\theta^*, y)} \\ &\xrightarrow{d} \frac{Z}{\mathcal{J}(\theta_0)} \\ &\sim N(0, \mathcal{J}(\theta_0)^{-1}), \end{aligned}$$

where $\mathcal{J}$ is known as the Fisher information matrix.

The model might be wrong. But under regularity conditions, the estimator will still converge to the θ' such that your posited model for the response is closest to the true model in KL divergence:

$$KL := \int \log \left(\frac{p_{\text{proposed}}(y)}{p_{\text{true}}(y)} \right) \cdot p_{\text{true}}(y) dy.$$